

Announcements – Jan 31

- HW 2 is due Friday at 9am.
- For the homework, be sure to assign problems to the pages when you turn it in. If you do not you will lose points.
- TA office hours will move to GS 856 starting today.
- SI sessions are Tue 6-7pm and Thu 2-3pm.
- Quiz 7 trivia – The company was Atari. A US company (originally) it was named after a Japanese term used in the game Go. (36% of you got this correct. It was the most common answer.)

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$\exists x P(x) \wedge \exists x Q(x)$ then $\exists x (P(x) \wedge Q(x))$

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- | | |
|---|------------------------------------|
| 1. $\exists x P(x) \wedge \exists x Q(x)$ | Premise |
| 2. $\exists x P(x)$ | Simplification from (1) |
| 3. $P(c)$ | Existential instantiation from (2) |
| 4. $\exists x Q(x)$ | Simplification from (1) |
| 5. $Q(c)$ | Existential instantiation from (4) |
| 6. $P(c) \wedge Q(c)$ | Conjunction of (3) and (5) |
| 7. $\exists x (P(x) \wedge Q(x))$ | Existential generalization |

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6. $P(c) \wedge Q(c)$	Conjunction of (3) and (5)
7. $\exists x (P(x) \wedge Q(x))$	Existential generalization

The problem is the c in 3 and in 5 are not necessarily the same element.



Combining Rules of Inference for Propositions and Quantified Statements

$$\forall x (P(x) \rightarrow Q(x))$$

$P(a)$, where a is a particular element in the domain

$$\therefore Q(a)$$

This is called *universal modus ponens*.



Combining Rules of Inference for Propositions and Quantified Statements

$$\forall x (P(x) \rightarrow Q(x))$$

$\neg Q(a)$, where a is a particular element in the domain

$$\therefore \neg P(a)$$

This is called *universal modus tollens*.

Specious Reasoning: The Bear Patrol

Homer: Ah, not a bear in sight. The Bear Patrol must be working like a charm!

Lisa: That's **specious reasoning**, Dad. [...] By your logic, I could claim that this rock keeps tigers away!

Homer: Uh-huh, and how does it work?

Lisa: It doesn't work. [...] It's just a stupid rock. [...] But I don't see any tigers around here, do you?

Homer: Lisa, I'd like to buy your rock.



From: **The Simpsons**, “Much Apu About Nothing”

(Season 7, Episode 151, Production Code 3F20)

Specious Reasoning: The Bear Patrol

Definition: Specious Reasoning

An unsupported or improperly constructed argument.
(That is, an unsound or invalid argument.)

Question: Where is the error in Homer's logic?

b: There are bears in Springfield

w: The Bear Patrol is working

First issue: What is Homer's argument?

Specious Reasoning: The Bear Patrol

Question: Where is the error in Homer's logic?

- (1) $\neg b$ (Given)
(2) $w \rightarrow \neg b$

(3) $\therefore w$

**Invalid
argument**

Fallacies

Three classic types:

1. Affirming the Conclusion

Example:

If you don't take the final, then you will fail the class. You failed the class. Therefore, you didn't take the final.

$$p \rightarrow q, q \mid \therefore p$$

Fallacies

Three classic types:

2. Denying the Hypothesis

Example:

If Margret is in Dallas, then she is in Texas. Margret isn't in Dallas. Therefore, she is not in Texas.

$$p \rightarrow q, \neg p \mid \therefore \neg q$$

Fallacies

Three classic types:

3. Begging the Question (a.k.a. Circular Reasoning)

Examples:

I'm not lying because I'm telling the truth.
It's like that because that's the way it is.

$\dots, p, \dots \mid \therefore p$

This is actually valid.
It's just not a good
"proof".

- This Slide Deck Covers Rosen 1.7 – 1.8

Some Definitions

- **Theorem:**
 - A statement that can be shown to be true. (proved)
- **Proof:**
 - A sound argument that establishes the truth of a theorem.
- **Axioms:**
 - Statements that are assumed to be true.
 - We can't prove anything without having some axioms defined.

- **Lemma:**
 - A theorem used to prove another theorem.
- **Corollary:**
 - A theorem that follows directly from another theorem.
- Which theorems are corollaries and lemmas is somewhat subjective.
- **Conjecture:**
 - A statement which is proposed to be true (hopefully based on some kind of evidence), but has not been proved.

Are Proofs Scary?

- Many students fear proofs and many students never get good at them.
- I believe those two things are related.
- I believe many students don't understand what proofs are.
- A proof does not have to be in a particular format or use a particular vocabulary or set of symbols.
- A proof is a communication of a sound argument that shows something is true.
- I like to use mathematical notation in proofs because it is concise and well defined. (I hate writing) However, a perfectly correct proof could write everything out in words.

Recall in our mythical island of Knights and Knaves, Knights always tell the truth and Knaves always lie. Given some people and their statements can you tell who is a Knight and who is a Knave? If not, can you tell anything?

You meet two characters, Phineas and Ferb.

1. Phineas says: “Ferb is a knave.”
2. Ferb says “Neither Phineas or I are knaves.”

Can you tell the types of both boys? Explain how.

Knights and Knaves

You meet two characters, Phineas and Ferb.

1. Phineas says: “Ferb is a knave.”
2. Ferb says “Neither Phineas or I are knaves.”

Claim: Phineas is a knight and Ferb is a knave.

Suppose Phineas is a knight. That means what he says is true, so Ferb is a knave.

Not done. I need to make sure there is no contradiction.

Ferb is a knave means what he says is false, which it is (since he is a knave).

Therefore, Phineas is a knight and Ferb is a knave works.

Still not done since we need to make sure there isn't another solution.

Suppose Phineas is a knave. That means what he says is false, so Ferb is a knight.

But if Ferb is a knight then what he says has to be true, but it's not since P is a knave.

So. Phineas is a knight and Ferb is a knave is the only solution.

Knights and Knaves

You meet two characters, Phineas and Ferb.

1. Phineas says: “Ferb is a knave.”
2. Ferb says “Neither Phineas or I are knaves.”

We could write the same proof more mathematically:

Claim: Phineas is a knight and Ferb is a knave.

Let P be Phineas is a Knight F be Ferb is a Knight and (n) mean statement n is true

Assume P

$\Rightarrow (1) \Rightarrow \bar{F} \Rightarrow \overline{(2)} = \bar{P} \vee \bar{F}$ which is true. /

Assume $\bar{P}$

$\Rightarrow \overline{(1)} \Rightarrow F \Rightarrow (2) \Rightarrow P \wedge F \Rightarrow \Leftarrow$ (since we assumed $\bar{P}$)

$\therefore$ Phineas is a knight and Ferb is a knave is the only solution. //

- The key to remember is a proof is just an explanation demonstrating the truth of some theorem (argument).
- A proof must be **complete**.
 - This means we can't leave any possibilities unaddressed.
 - For example, if I were trying to prove every rational number has a multiplicative inverse, i.e.:
 - $\forall x \in \mathbb{Q} \exists y \in \mathbb{Q} (x \cdot y = 1)$
 - And I say for any x choose y to be $1/x$, what case am I leaving out?
 - $x = 0$